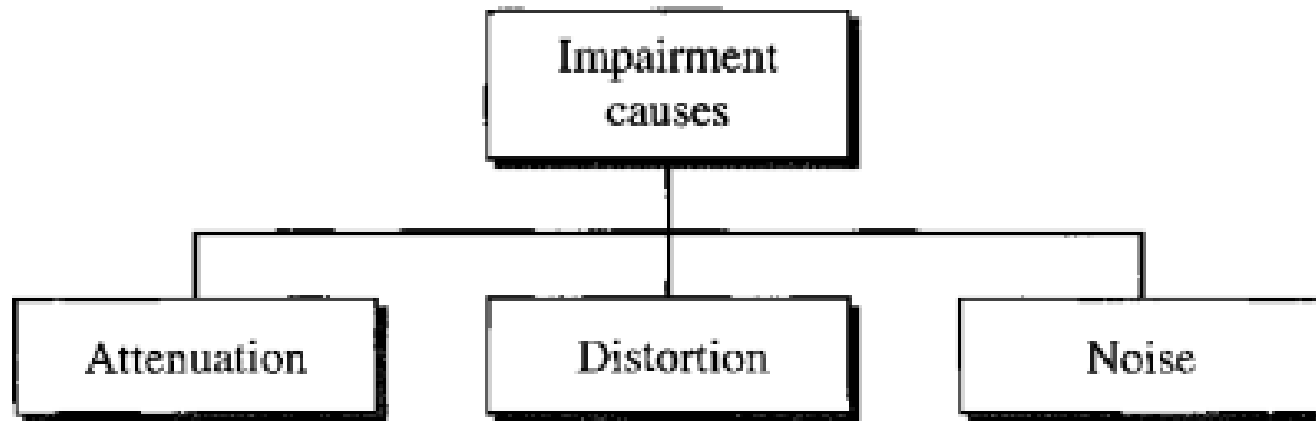


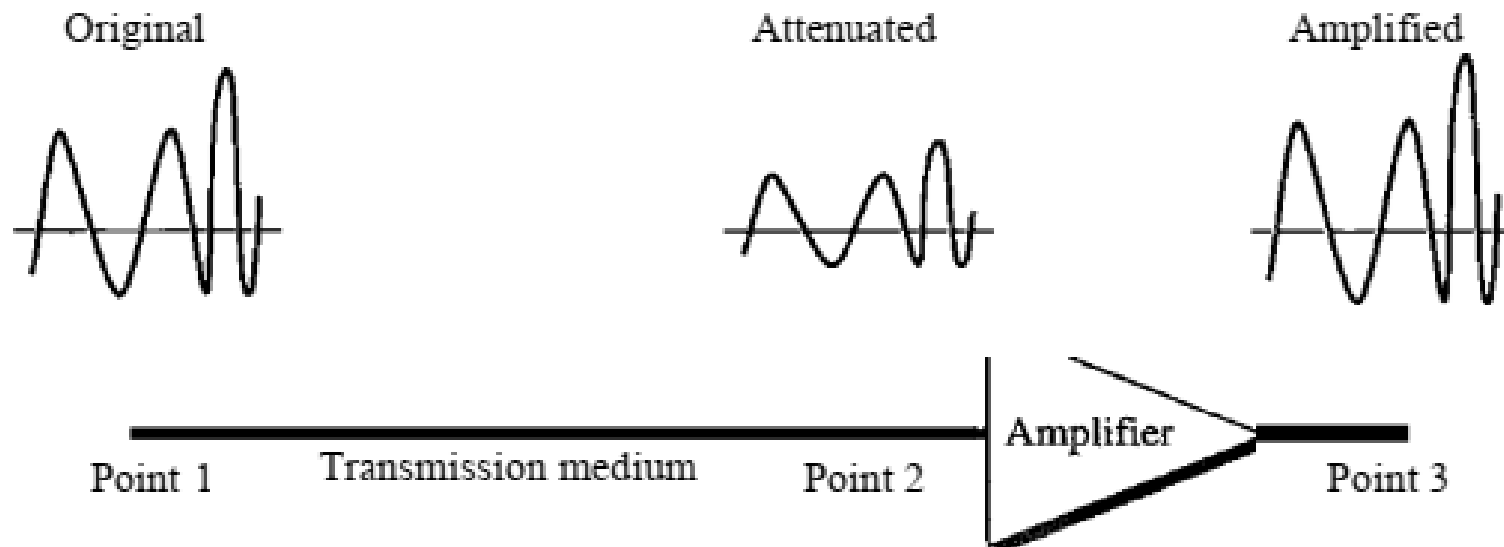
Transmission Impairment

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise



Attenuation

- ❑ Attenuation means a loss of energy.
- ❑ When a signal (simple or composite) travels through a medium, it loses energy due to the resistance of the medium.
- ❑ To compensate for this loss, amplifiers are used to strengthen (amplify) the signal.



Attenuation

- ❑ Engineers use the decibel (dB) to indicate whether a signal has lost or gained strength.
- ❑ The decibel measures the relative strength between two signals.
- ❑ It can also compare the strength of one signal at two different points.

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

Variables P_1 and P_2 are the powers of a signal at points 1 and 2, respectively. Note that power is proportional to the square of the voltage; So, the formula is $\text{dB} = 20 \log_{10} (V_2/V_1)$.

Attenuation

Example 1

A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10 \cdot P_1$. In this case, the amplification (gain of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1} = 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

Attenuation

Example 2

The loss in a cable is usually expressed in decibels per kilometer (dB/km). If a cable has a loss of -0.3 dB/km and the signal power at the beginning of the cable is 2 mW, what will be the signal power after the signal has traveled 5 km through the cable?

Solution: Given

Loss = **-0.3 dB/km**; Distance = **5 km**; Initial power $P_1 = 2 \text{ mW}$

Total loss = $(-0.3 \text{ dB/km}) \times 5 \text{ km} = -1.5 \text{ dB}$

$$\text{dB} = 10 \log_{10} \left(\frac{P_2}{P_1} \right) = -1.5$$

$$\frac{P_2}{P_1} = 10^{-0.15} = 0.71$$

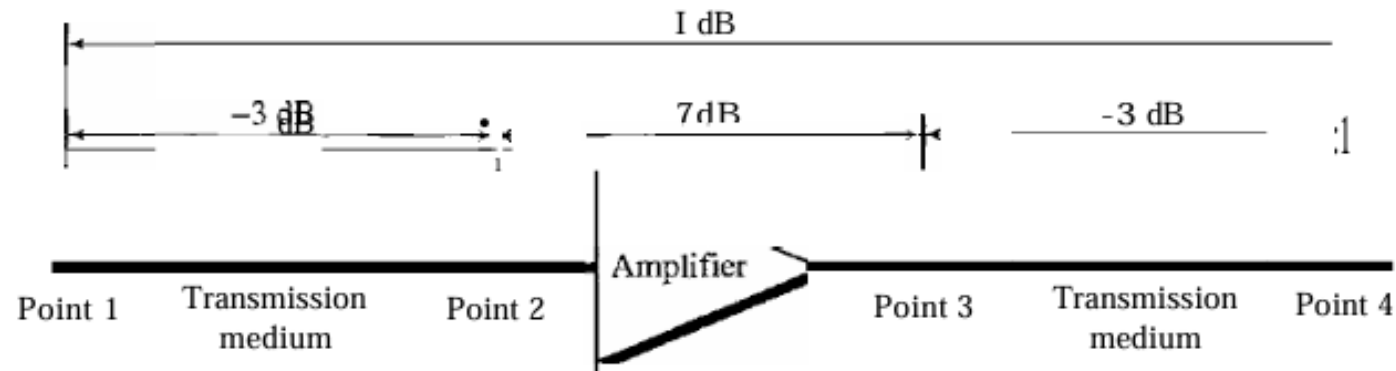
$$P_2 = 0.71 P_1 = 0.7 \times 2 = 1.4 \text{ mW}$$

Attenuation

Example 3

Engineers use the **decibel (dB)** to measure changes in signal strength because dB values can be **added or subtracted** when analyzing multiple cascading points.

If a signal travels from point 1 to point 4—attenuated from 1 to 2, amplified from 2 to 3, and attenuated again from 3 to 4—the overall effect can be found simply by **adding the dB values** for each section.

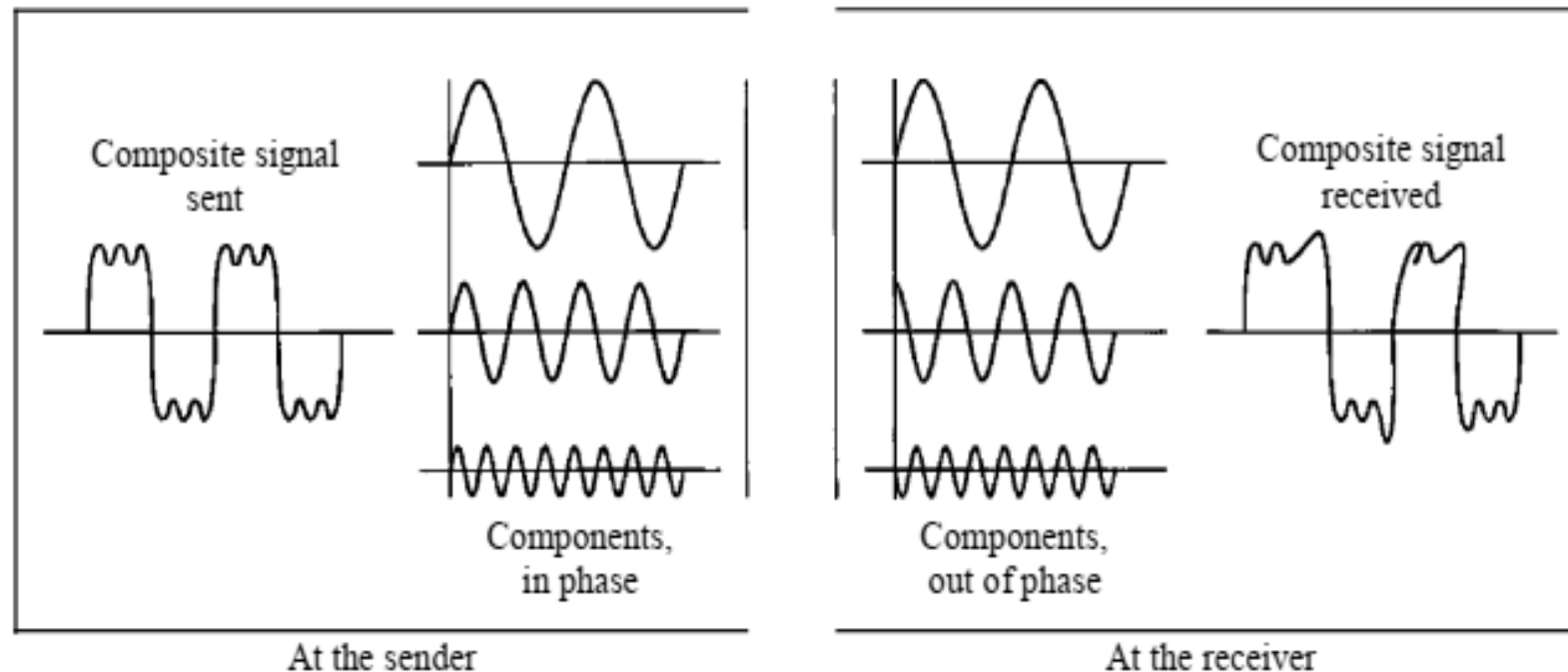


In this case, the decibel value can be calculated as

$$\text{dB} = -3 + 7 - 3 = +1$$

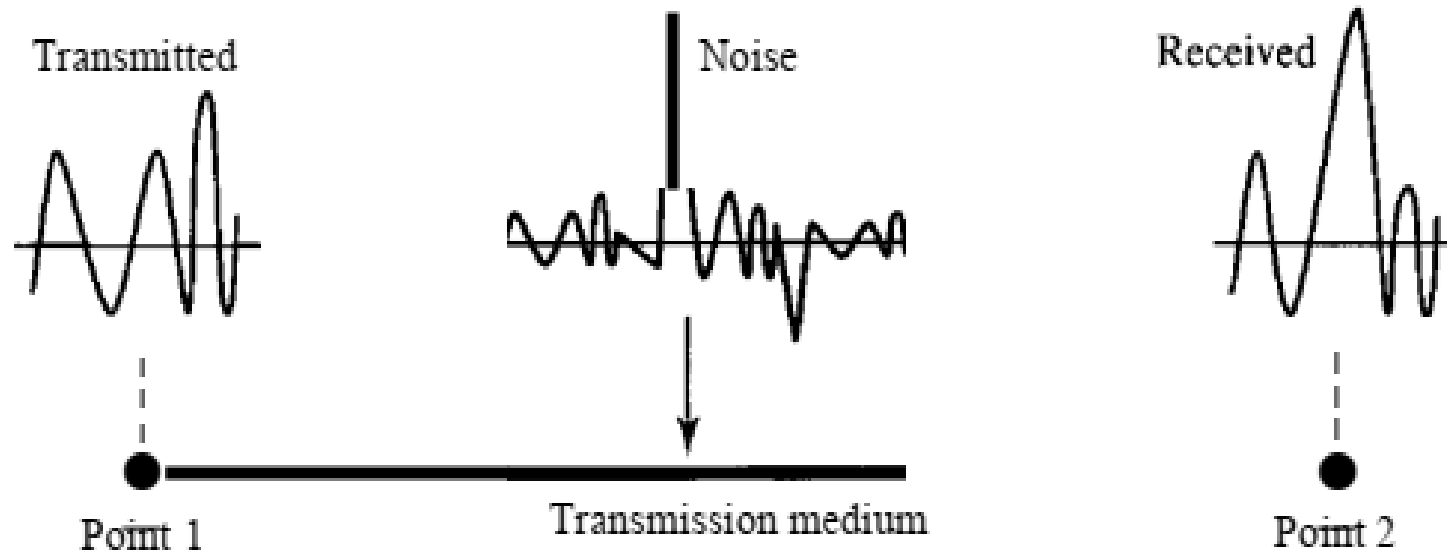
Distortion

Distortion means that the signal changes its form or shape. Distortion can occur in a composite signal made of different frequencies. Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration. Figure below shows the effect of distortion on a composite signal.



Noise

Noise in a communication system is basically undesirable or unwanted signals that get randomly added to the actual information carrying signal. Induced noise comes from sources such as motors and appliances. These devices act as a sending antenna, and the transmission medium acts as the receiving antenna. The random motion of the electrons is the reason for the thermal noise received by the conductor.



Noise

Signal-to-Noise Ratio (SNR)

We need to know the ratio of the signal power to the noise power. The signal-to-noise ratio is defined as

$$\text{SNR} = \frac{\text{Average signal Power}}{\text{Average noise Power}}$$

We need to consider the average signal power and the average noise power because these may change with time. SNR is actually the ratio of what is wanted (signal) to what is not wanted (noise). A high SNR means the signal is less corrupted by noise; a low SNR means the signal is more corrupted by noise. Because SNR is the ratio of two powers, it is often described in decibel units, SNR(dB), defined as

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right)$$

Noise

Example 3

The power of a signal is 10 mW and the power of the noise is 1 μ W; what are the values of SNR and SNR(dB)?

Solution:

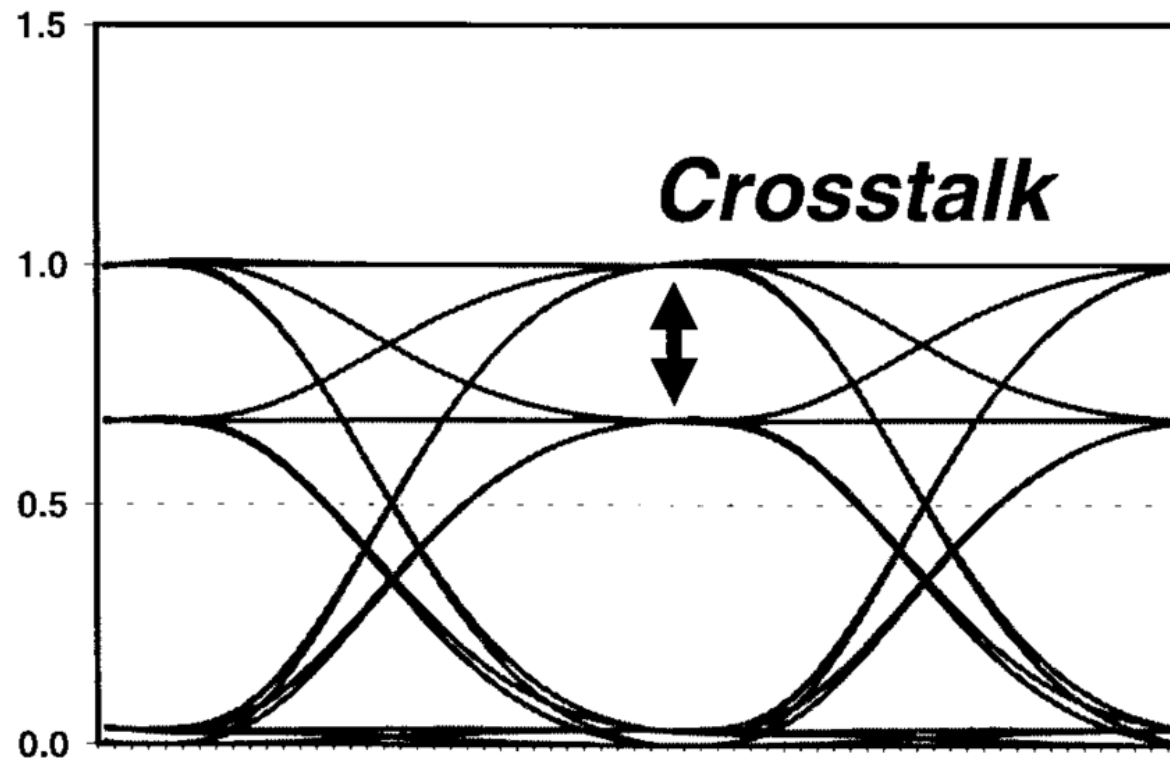
The values of SNR and SNR_{dB} can be calculated as follows:

$$\text{SNR} = \frac{10.000 \text{ mW}}{1 \mu\text{W}} = 10.000$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} 10.000 = 10 \log_{10} 10^4 = 40$$

Crosstalk

In electronics, crosstalk is any phenomenon by which a signal transmitted on one circuit or channel of a transmission system creates an undesired effect in another circuit or channel. Crosstalk is usually caused by undesired capacitive, inductive, or conductive coupling from one circuit or channel to another.



Data Rate Limit

A very important consideration in data communications is how fast we can send data, in bits per second (bps) over a channel. Data rate depends on three factors:

- The bandwidth available
- The level of the signals we use
- The quality of the channel (the level of noise)

Two theoretical formulas were developed to calculate the data rate:

- one by Nyquist for a noiseless channel
- another by Shannon for a noisy channel

Data Rate Limit

Noiseless Channel: Nyquist Bit Rate

It states that to reconstruct a continuous analog signal from its sampled version accurately, the sampling rate must be at least twice the highest frequency present in the signal.

The Formula for Nyquist Sampling Theorem can be given as

$$f_s \geq 2f_m$$

Where,

f_s refers to frequency signal

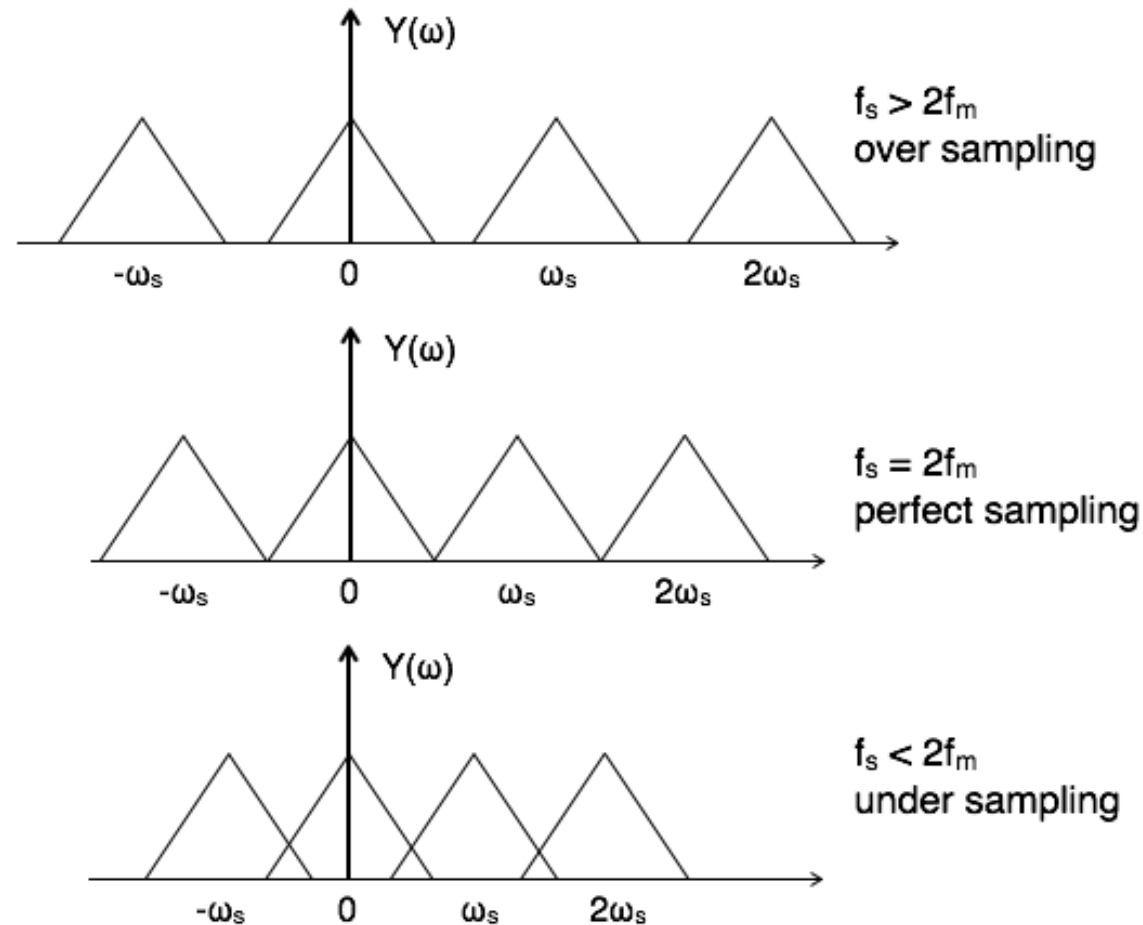
f_m refers to maximum frequency

This ensures that there are enough samples taken per unit of time to capture all the details of the original waveform without introducing aliasing, which can cause distortion or artifacts in the reconstructed signal.

Data Rate Limit

Noiseless Channel: Nyquist Bit Rate

If the waveform is under sampled (when. $f_s < f_m$) then the signal at the output of the filter will be different from the original signal spectrum This results in aliasing.



Data Rate Limit

Noiseless Channel: Nyquist Bit Rate

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

$$\text{Bit rate} = 2 * \text{Bandwidth} * \log_2 L$$

Here L is the number of signal level used to represent data

N:B: Increasing the levels of a signal may reduce the reliability of the system.

Channel Capacity

Consider noise free channel

- ▶ Nyquist rate: $C = 2B$
 - Given bandwidth B , highest signal rate is $2B$
 - If rate of signal transmission is $2B$, then we can carry signal with frequencies no more than B
 - Given binary signal, data rate supported by B Hz is $2B$ bps
- ▶ Can be increased by using M signal levels
- ▶ $C = 2B \log_2 M$
- ▶ Limited by noise and other impairments

Data Rate Limit

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\begin{aligned}\text{Bit rate} &= 2 * \text{Bandwidth} * \log_2 L \\ &= 2 * 3000 * \log_2 2 \\ &= 6000 \text{ bps}\end{aligned}$$

Data Rate Limit

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as

$$\begin{aligned}\text{Bit rate} &= 2 * \text{Bandwidth} * \log_2 L \\ &= 2 * 3000 * \log_2 4 \\ &= 12,000 \text{ bps}\end{aligned}$$

Data Rate Limit

Example 6

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

$$\text{Bit rate} = 2 * \text{Bandwidth} * \log_2 L$$

$$265,000 = 2 * 20,000 * \log_2 L$$

$$\log_2 L = 6.625$$

$$L = 2^{6.625}$$

$$L = 98.7 \text{ levels}$$

Data Rate Limit

Noisy Channel: Shannon Capacity

In reality, we cannot have a noiseless channel; the channel is always noisy. To determine the theoretical highest data rate for a noisy channel:

$$\text{Capacity} = \text{Bandwidth} * \log_2(1 + \text{SNR})$$

Note that in the Shannon formula there is no indication of the signal level, which means that no matter how many levels we have, we cannot achieve a data rate higher than the capacity of the channel. In other words, the formula defines a characteristic of the channel, not the method of transmission.

Channel Capacity

- ❑ **Consider data rate, noise and error rate**
- ❑ Faster data rate shortens each bit so burst of noise affects more bits
 - At given noise level, high data rate means higher error rate
- ❑ Signal to noise ration (in decibels)
- ❑ $\text{SNR}_{\text{db}} = 10 \log_{10} (\text{signal/noise})$
- ❑ Capacity $C = B \log_2(1 + \text{SNR})$
- ❑ This is error free capacity

Data Rate Limit

Example 7

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$\begin{aligned}\text{Capacity} &= \text{Bandwidth} * \log_2(1 + \text{SNR}) \\ &= B * \log_2(1 + 0) \\ &= B * \log_2(1) \\ &= 0\end{aligned}$$

Data Rate Limit

Example 8

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$\begin{aligned}\text{Capacity} &= \text{Bandwidth} * \log_2(1 + \text{SNR}) \\ &= 3000 * \log_2(1 + 3162) \\ &= 3000 * 11.62 \\ &= 34,860 \text{ bps}\end{aligned}$$

Data Rate Limit

Example 9

The signal-to-noise ratio is often given in decibels. Assume that SNR = 36 dB and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR} \quad \Rightarrow \quad \text{SNR} = 10^{\text{SNR}_{\text{dB}}/10} \quad \Rightarrow \quad \text{SNR} = 10^{3.6} = 3981$$
$$C = B \log_2 (1 + \text{SNR}) = 2 \times 10^6 \times \log_2 3982 = 24 \text{ Mbps}$$

Data Rate Limit

Using Both Limits

In practice, we need to use both methods to find the limits and signal levels. Let us show this with an example

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

First, we use the Shannon formula to find the upper limit.

$$C = B \log_2 (1 + \text{SNR}) = 10^6 \log_2 (1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \implies L = 4$$

Reference

Data Communication and Networking
-Behrouz A Forouzan
(3rd Chapter)